

Exploratory Data Analysis

Univariate & Multivariate Analysis Techniques

This report documents a practical exploration of data visualization techniques used to understand individual variables and the relationships between them, applied across four well-known datasets: Titanic, Tips, Flights, and Iris.

1. Introduction & Objective

Exploratory Data Analysis (EDA) is the process of investigating a dataset to summarize its main characteristics, often using visual methods, before applying any modeling. Choosing the right plot for the right data type is a foundational skill — the wrong chart can hide patterns or, worse, suggest patterns that aren't there.

The objective of this project is to practice and demonstrate the most common univariate and multivariate visualization techniques in Python, using pandas, matplotlib, and seaborn.

2. Datasets Used

Dataset	Source	Used For
Titanic	titanic.csv (local file)	Categorical & numerical univariate analysis, bivariate analysis
Tips	seaborn.load_dataset('tips')	Scatter plot (numerical vs numerical)
Flights	seaborn.load_dataset('flights')	Pivot table & cluster map
Iris	seaborn.load_dataset('iris')	Pair plot

The Titanic dataset — containing passenger demographics, ticket class, fare, and survival outcome — served as the primary dataset for both univariate and bivariate analysis, since it naturally contains a mix of categorical and numerical variables.

3. Univariate Analysis

Univariate analysis examines one variable in isolation to understand its distribution and spread.

3.1 Categorical Data

- **Count plot** — visualized the frequency of each category in the 'Sex' column.
- **Pie chart** — showed the proportion of passengers in each ticket class (Pclass).

3.2 Numerical Data

- **Histogram** — plotted the distribution of passenger 'Age' across 50 bins.
- **Distribution plot (displot)** — showed the Age distribution with a smoothed density curve (KDE).
- **Box plot** — summarized the Age distribution's median, quartiles, and outliers.
- Basic summary statistics (minimum, maximum, and mean) were also computed for the Age column.

4. Multivariate Analysis

Multivariate analysis examines the relationship between two or more variables, which is where interesting patterns and correlations tend to emerge.

Relationship Type	Technique	Example Used
Numerical vs Numerical	Scatter plot	Tips: total_bill vs tip, colored by sex and smoker status
Categorical vs Categorical	Bar plot	Titanic: Fare by Pclass, split by Sex
Numerical vs Categorical	Box plot	Titanic: Age by Sex, split by Survived
Numerical vs Categorical	Distribution plot	Titanic: Age distribution compared between survivors and non-survivors
Multiple numerical variables	Pair plot	Iris: all feature pairs, colored by species
Aggregated / pivoted data	Pivot table + Cluster map	Flights: passenger counts by month and year

4.1 Key Observation — Survival Analysis

Comparing the Age distribution between passengers who survived and those who did not (using overlaid distribution plots) is a classic example of how a single visualization can suggest a hypothesis — for instance, whether younger passengers had a higher chance of survival — that can then be tested more formally with statistics or modeling.

5. Tools & Libraries

- **pandas** — data loading and manipulation
- **numpy** — numerical operations
- **matplotlib** — base plotting (histograms)
- **seaborn** — statistical visualizations (count plots, box plots, pair plots, cluster maps, etc.)

6. Conclusion

This project served as a practical reference for matching visualization techniques to data types: count plots and pie charts for categorical data, histograms and box plots for numerical data, and scatter plots, bar plots, and pair plots for exploring relationships between variables. These techniques form the groundwork for any deeper statistical analysis or machine learning work that follows.